

# Principles of Macroeconomics

Personal Notes

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# 1 Introduction to 14.02 Principles of Macroeconomics

Macroeconomics studies the behavior of an economy as a whole. Its objects are aggregates and the links among them: the United States or China rather than a single household or firm, the inflation rate rather than one good's price, the unemployment rate rather than one worker's employment outcome, and the exchange rate rather than the relative price of two particular goods.

## 1.1 Why Macro Is Not Just “Lots of Micro”

An economy is made up of households, firms, and individual markets, so it is tempting to treat macroeconomics as the sum of many microeconomic decisions. That construction misses the central difficulty. Once many agents interact, their beliefs, prices, and decisions must fit together in an equilibrium; the aggregate system has relationships that are hard to see by studying an individual component in isolation.

Macroeconomists therefore use deliberately simplified models. The goal is not to reproduce every institutional detail, but to retain the mechanism that matters for the question at hand. This is a judgment as well as a technical exercise: a model should be simple enough to reveal the essential aggregate relationship without becoming so simple that it explains the wrong thing.

*Note: The course uses modest mathematics. A concept may be subtle, but the aim is to understand the economic forces, not to turn the introductory course into a PhD-level modelling sequence.*

## 1.2 What the Course Is Meant to Make Legible

The practical objective is economic literacy. By the end of the course, a student should be able to read an IMF *World Economic Outlook* or a financial newspaper, identify the macroeconomic variables at issue, and evaluate the proposed explanation rather than merely accept it. The course is therefore organized around recurring current events. As the vocabulary and models develop, the same news story can be revisited with a more precise account of its causes and consequences.

Formal definitions come later. This opening lecture instead introduces the kind of evidence macroeconomists use and the questions they ask of it:

- Why do inflation and wage growth move together, and when should that connection concern policymakers?
- Why does unemployment rise in recessions and recover at different speeds after different shocks?
- How can a central bank influence spending, investment, and asset values through the interest rate?

- Why can the same worldwide episode have a common core but different country-specific amplifiers?

### 1.3 A Macroeconomic Snapshot: Inflation after COVID-19

The lecture uses the early-2023 U.S. economy as a motivating case. The unemployment rate rises during recessions; the Great Recession was followed by a long labor-market recovery, while the sharp COVID-19 disruption was followed by a much faster rebound. By the period discussed in the lecture, unemployment was exceptionally low and wage growth was rapid, especially in accommodation and food services.

Those facts are welcome from an individual worker's perspective, but macro analysis asks about their joint implications. Wage growth and inflation tend to be strongly related. The usual inflation objective for large advanced economies is roughly 2% per year, whereas the lecture describes readings around 6.5% to 8% in the United States. Inflation had begun to ease, but remained far above that benchmark.

**Definition:** *Inflation* is the rate at which the overall price level rises. It is an aggregate concept: it is not the price increase of one product. A precise measurement definition is deferred to the next lecture.

The post-COVID inflation episode was broad, not uniquely American. Many economies that had previously experienced inflation near or below 2% saw it rise substantially. The shared backdrop was an unusually strong recovery; local conditions then added their own effects. In Europe, for example, the war in Ukraine and the associated energy-price shock intensified inflation.

***Note:** Numbers in this section are the lecture's contemporaneous examples, not current estimates. Their role is to show how macroeconomic data become a policy problem.*

### 1.4 Interest Rates, Aggregate Demand, and Financial Markets

The central bank's principal instrument for controlling inflation is the interest rate. Lower interest rates make borrowing less expensive, which tends to support consumption and firms' investment. Higher rates work in the opposite direction, restraining demand. This trade-off explains the rapid rate increases undertaken once policymakers judged inflation to be too high.

Monetary policy also reaches financial markets. In the lecture's example, the stock market fell sharply when COVID-19 first appeared, then rose well above its pre-pandemic level while monetary policy was very loose. Later rate hikes were associated with a broad decline in equity values. The point is not that one chart proves a complete causal story, but that interest rates change the discounting and financing conditions relevant to many assets.

Financial prices can move immediately when new information arrives because they embody expectations about the future. An employment release, for example, can change investors' forecasts of inflation and future monetary policy before the stock market has even opened. Macroeconomics supplies the framework for tracing those links rather than treating each data release as an isolated headline.

## 1.5 Takeaways

- Macro studies aggregate outcomes and the interactions that connect them; it needs models tailored to those interactions.
- Good macro models make disciplined simplifications so that the key mechanism is visible.
- Inflation, unemployment, wage growth, interest rates, and asset prices are jointly informative, not separate scorecards.
- Current events are both an application of macroeconomic tools and a source of questions that motivate building those tools.

*Source: MIT 14.02 Principles of Macroeconomics, Spring 2023, Lecture 1 video by Ricardo J. Caballero.*